

The effect of acute exercise on cognitive and motor inhibition – Does fitness moderate this effect?

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ABSTRACT

Background: Given the extensive evidence on improvements in cognitive inhibition immediately following exercise, and the literature indicating that cognitive and motor inhibitory functions are mediated by overlapping brain networks, the aim of this study was to assess, for the first time, the effect of moderate intensity acute aerobic exercise on multi-limb motor inhibition, as compared to cognitive inhibition.

Method: Participants were 36 healthy adults aged 40–60 years old (mean age 46.8 ± 5.7), who were randomly assigned to experimental or control groups. One-to-two weeks following baseline assessment, participants were asked to perform a three-limb (3-Limb) inhibition task and a vocal version of the Stroop before and after either acute moderate-intense aerobic exercise (experimental group) or rest (control).

Results: Similar rates of improvement were observed among both groups from baseline to the pre-test. Conversely, a meaningful, yet non-significant trend was seen among the experimental group in their pretest to posttest improvement in both cognitive and motor tasks. In addition, exploratory analysis revealed significant group differences in favor of the experimental group among highly fit participants on the 3-Limb task. A significant correlation was indicated between the inhibition conditions, i.e., choice in the motor inhibition and color/word (incongruent) in the cognitive inhibition, especially in the improvement observed following the exercise.

Discussion: Moderate-intensity acute aerobic exercise is a potential stimulator of both multi-limb motor inhibition and cognitive inhibition. It appears that high-fit participants benefit from exercise more than low-fit people. Additionally, performance on behavioral tasks that represent motor and cognitive inhibition is related. This observation suggests that fitness levels and acute exercise contribute to the coupling between cognitive and motor inhibition. Neuroimaging methods would allow examining brain-behavior associations of exercise-induced changes in the brain.

1. Introduction

Cognitive control, also referred to as executive control or executive functioning (Aron, 2007; Diamond, 2013), refers to a family of top-down mental processes needed for concentrating and paying attention when an automatic mental response is impossible or undesired (Diamond, 2013). Inhibition is a basic element of all executive functions, and is considered a core process in cognitive control. It allows the suppression of an irrelevant response, stimulus, or memory (Aron, 2007), and involves the ability to shift or ignore a strong internal predisposition or external stimulus as a means for performing the correct response (Aron,

2007, 2011; Diamond, 2013).

Inhibition is regulated through various cortical and subcortical networks and is considered a key mechanism underlying cognitive [e.g., (Nielson et al., 2002)] and motor performance [e.g., (Fujiyama et al., 2016)]. For cognitive control, inhibition can be conceptualized as a process that blocks the spread of neural activity to or from nearby regions that are not relevant to a task at hand (Talelli et al., 2008). For motor control, inhibition is required during withdrawal, cancellation, or selection of voluntary movements (Mirabella, 2014; Stinear et al., 2009), the prevention of prepotent responses [e.g., (Duque & Ivry, 2009)], or attraction towards preferred patterns of movements [e.g.,

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(Heuninckx et al., 2004, pp. P225–P232)].

From a network point of view, cognitive and motor inhibitory functions are mediated by overlapping brain networks comprising the prefrontal cortices and basal ganglia (Aron, 2007; Mirabella, 2014). More specifically, inhibition is mediated primarily by a widespread prefrontal network including the right middle and right inferior frontal gyri, the pre-supplementary motor area, the left-dorsolateral prefrontal cortex, and the basal ganglia (Depue et al., 2016; Fujiyama et al., 2016; Levin et al., 2014; Tan et al., 2019).

In contrast to cognitive tasks such as Stroop, where inhibition is thought to be largely subserved by the dorsal-frontal pathways [e.g., (Hung et al., 2018; Wolf et al., 2014)], motor inhibition during stop-signal or go-no-go tasks is thought to be implemented primarily by the fronto-striatal pathways (Aron & Poldrack, 2006; Hung et al., 2018; Weerasekera et al., 2020). Both pathways interact with the anterior insula and the dorsal anterior cingulate cortex, which are regarded as principal brain regions involved in the regulation of “top-down, attentional-driven cognitive control processes” [(Hung et al., 2018), p. 4075].

The effect of normal aging on executive functions including cognitive and motor inhibition has been studied extensively. For example, age-related changes in neural activation (assessed by functional magnetic resonance imaging - fMRI) across a broad spectrum of executive function tasks, and age-related changes in gray matter (using voxel-based morphometry - VBM) have been reported during adult life suggesting that these changes occur at various points in adult life. Importantly, these changes often occur at mean age of 40–50 years old or even at a younger age (Di et al., 2014). As individuals age beyond a certain point, they experience significant gray matter reductions in the prefrontal cortex, as well as age-related deficits in the functional hyperactivation aspect of executive function (Di et al., 2014).

A large body of evidence shows that in comparison to younger adults, older adults are less able to suppress dominant responses or movement paradigms [e.g., (Levin et al., 2014; Rey-Mermet & Gade, 2018)], or effectively apply top-down suppression of irrelevant information [e.g., (Chadick et al., 2014)]. On a behavioral level, the effect of age on inhibitory control increases with task complexity, suggesting that older adults may be less efficient than younger adults at performing multi-limb coordination tasks with their four limbs as previously shown by Boisgontier et al. (Boisgontier et al., 2014, 2016). In addition, older adults are known to shift the speed/accuracy trade-off towards accuracy or increase cognitive control to improve response selection and response inhibition (Coelho et al., 2021; Seer et al., 2021). However, limited attention has been given to exploring how age affects the interaction between cognitive and motor inhibition.

While aging negatively affects brain structure and function (Boisgontier et al., 2016; Bonifazi et al., 2018; King et al., 2018; Wolf et al., 2014), there is evidence that such alterations can be delayed or even reversed, and that certain skills can be revived through training-induced neuroplastic changes [e.g., (Erickson et al., 2007)]. One intervention associated with behavioral and neuroplastic changes is physical exercise, with chronic exercise training being shown to be positively associated with cognitive inhibition in older adults [e.g., (Boucard et al., 2012; Ishihara et al., 2021a; Oliveira et al., 2021)]; for a review see (Chen et al., 2020)].

Complementing the above-mentioned literature on chronic exercise, a rapidly growing number of investigations have focused on the effects of a single session of exercise (i.e., acute exercise) on cognitive performance, reporting improvements in various aspects of cognition [for reviews, see: (Ai et al., 2021; Basso & Suzuki, 2017; Chang et al., 2012; Hsieh et al., 2021; Ishihara et al., 2021b; Levin et al., 2021; McSween et al., 2019)].

A possible mechanism that may underlie the beneficial effects of acute cardiovascular exercise on inhibitory control or other executive functions is energetic arousal – increases in neural activation or general physiological arousal measured by heart rate, oxygen uptake, or other

biological indices (Lambourne & Tomporowski, 2010). Specifically, exercise may have a priming effect on the central nervous system by transiently increasing blood flow to prefrontal or subcortical regions that regulate system inhibition such as prefrontal-basal ganglia pathways (Levin et al., 2021). Another commonly mentioned mediator of acute exercise and cognition is the increase in the levels of some nerve growth factors, such as the brain derived neurotrophic factor (BDNF) (Knaepen et al., 2010). The transitory effect of an acute bout of exercise has also been attributed to a change in catecholamine concentrations in the brain, as a result of cortisol release induced by exercise (McMorris, 2016).

The exercise-induced arousal or the elevated BDNF level or the change in catecholamine concentrations are expected to be maintained for a short period following exercise, and cognitive performance is also expected to be facilitated during this time.

Specifically, improvements in cognitive inhibitory control immediately following exercise have been observed in young adults (Dora et al., 2021; Endo et al., 2013; Hillman et al., 2003), middle-aged adults (Chang et al., 2019; Netz et al., 2016; Nouchi et al., 2020), and healthy older adults (Abe et al., 2018; Hyodo et al., 2012; Kamijo et al., 2009). The main findings show that a single bout of exercise can enhance inhibitory control [for reviews see (Ishihara et al., 2021b; Oberste et al., 2019)].

In the present study we focus on the effects of acute aerobic exercise on motor inhibition in middle-aged adults. To the best of our knowledge, motor inhibition performance tasks involving multi-limb task complexity (4-limb coordination) have not been studied in relation to acute aerobic exercise. For comparison purposes and for assessing the interaction between cognitive and motor inhibition we also administered a cognitive inhibition test (Stroop, 1935).

For assessing multi-limb task complexity, we based our study on the model of coupling/decoupling interactions (Boisgontier et al., 2014). According to this model the brain is considered a highly interconnected neural space with different gradients of coupling and decoupling interactions. Activation in one area may therefore spread to other areas to different extents, which requires recruitment of inhibitory mechanisms to enable selective movement generation. This model suggests that the default mode of the motor system is a basic coupling between the limbs while in other movements, a decoupling is required and thus calls for inhibitory interactions. More specifically, it was postulated that overall processing complexity of a task might be determined by the recruitment (the more effector-specific networks recruited the lower the performance) and selection (the coupling/decoupling interactions involved in a given coordination mode determine performance) principles. The simple reaction time (RT) represents the recruitment complexity, while the choice RT involves selection complexity in addition to the recruitment complexity, thus requires more inhibition effort.

In summary, the aim of this study was to assess, for the first time, the effect of moderate intensity acute aerobic exercise on multi-limb motor inhibition, as compared to cognitive inhibition. Given the evidence of improvements in cognitive inhibition following acute exercise, and given that cognitive and motor inhibitory functions are mediated by overlapping brain networks, we hypothesized that: (1) the inhibited response, but not the simple response – in both the cognitive and the motor tasks will be altered by the exercise; and (2) motor and cognitive inhibition will be related.

2. Method

2.1. Participants

Statistical power was calculated using G*Power (Faul et al., 2009) to find an interaction in a two-way ANOVA based on the following criteria: Cohen's $f = 0.25$, $\alpha = .05$, two groups, two measurements, correlation between repeated measures = .05, nonsphericity correction = 1 in a Group $\times$ Test interaction of a two-way analysis of variance (ANOVA).

The analysis showed that 34 participants were required to achieve 80% power. In the current study, we were able to recruit 36 healthy and physically active volunteers (21 women, 15 men), aged 40–60 (mean age 46.8 ± 5.7), who participated in the study and provided 83% power (see more details on participants in Table 1). The study was approved by the ethics committee of The Academic College at Wingate and all the participants signed an informed consent form prior to participation. The participants were assigned to experimental and control groups while controlling for gender using permuted block randomization calculated in R (R Core Team, R, 2020).

2.2. Clinical and demographic assessment

Assessment included demographic information (age, marital status, birthplace, employment, education, smoking status, height and weight, and mother tongue); habitual physical activity (average weekly time spent on aerobic/non-aerobic exercise), cognitive evaluation (Folstein et al., 1975), computer use, and interview with a physician (Table 1).

2.3. Determining individual moderate intensity level: predicted peak VO_2

A physician administered a graded maximal exercise test on a motorized treadmill to predict peak VO_2 and to measure maximal heart rate (HR_{max}). Based on this test, we calculated the exercise intensity as 60% of Heart Rate Reserve (HRR) for each individual participant. The test also served as an exclusion criterion. However, no abnormal cardiac symptoms were detected among the participants. For the duration of the test, the electrocardiogram (ECG) and heart rate (HR) were continuously monitored, using a 12-lead ECG, and a sphygmomanometer measured the participants' blood pressure. The test commenced with 2–5 min of warm-up. Based on the Bruce protocol (American College of Sports Medicine (ACSM), 2014), an initial speed of $3.2 \text{ km} \cdot \text{hr}^{-1}$ with a gradient of zero was determined. The gradient was increased by 2.5% every 2 min, until a negative symptom appeared (e.g., dizziness, breathlessness, or changes in the ECG), HR_{max} was reached, or the participant reached his/her limit of tolerance. The following equation was used to estimate peak VO_2 from the last stage of the test: $VO_2 (\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}) = 0.1 \cdot (\text{final speed}) + 1.8 \cdot (\text{final speed}) \cdot (\text{final fractional grade}) + 3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$

Table 1

Participants' demographic, physical, and clinical characteristics (means $\pm$ SD when relevant).

Variables	Gender	Experimental Group	Control Group
N (women, men)		20 (11,9)	16 (10,6)
Age (years)		45.62 ± 5.41	48.25 ± 5.98
Height (cm)	Female	162 ± 6	163 ± 6
	Male	179 ± 6	175 ± 4
Weight (kg)	Female	66.82 ± 15.73	69.27 ± 15.20
	Male	81.89 ± 13.95	75.00 ± 5.58
BMI ($\text{kg} \cdot \text{m}^{-2}$)	Female	24.91 ± 4.22	26.16 ± 5.53
	Male	25.48 ± 2.78	24.56 ± 2.60
Live in central Israel (%)		85	93.75
Married (%)		90	75
Smoke (N)		2	1
Education (years)		16.50 ± 4.02	16.62 ± 3.36
Computer use ($\text{hr} \cdot \text{week}^{-1}$)		23.77 ± 17.63	31.62 ± 15.99
MMSE		29.45 ± 0.83	29.56 ± 0.73
HR _{rest} ($\text{beats} \cdot \text{min}^{-1}$)	Female	77.63 ± 12.58	85.70 ± 14.20
	Male	81.55 ± 12.37	72.67 ± 20.00
HR _{max} ($\text{beats} \cdot \text{min}^{-1}$)	Female	171.09 ± 11.54	171.10 ± 6.57
	Male	176.00 ± 13.98	172.17 ± 11.51
METs	Female	9.94 ± 1.37	9.87 ± 2.20
	Male	12.61 ± 2.55	12.17 ± 2.05
Predicted $VO_{2\text{max}}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	Female	34.80 ± 4.79	34.56 ± 7.70
	Male	44.16 ± 8.93	42.61 ± 7.18

BMI = Body Mass Index, HR_{max} = Maximal heart rate, HR_{rest} = Heart rate at rest, METs = Metabolic Equivalents, MMSE = Mini-Mental State Examination.

(American College of Sports Medicine (ACSM), 2014).

2.4. Motor performance assessment

2.4.1. Full battery for habituation

The participants were seated in front of a computer screen, with their forearms resting on a table and their fingers and forefeet placed on tablets with capacitance sensors (Fig. 1). Four squares representing the four limb segments were presented on the screen - the right and left upper squares represent the right and left hands, and the right and left lower squares represent the right and left feet. Once all 4 limb segments had contacted the tablets, some of the squares on the screen turned blue. In response to this stimulus, participants had to release contact with the corresponding tablets as quickly as possible by simultaneously lifting the corresponding limb segment(s).

The full battery included all 15 possible coordination modes for limbs recruitment in simple and in choice reaction time (RT) conditions: 4 modes of 1-limb, 2 modes of 2-homologous upper/lower-limbs, 2 modes of 2-ipsilateral right/left-limbs, 2 modes of 2-diagonal limbs, 4 modes of 3-Limbs, and one 4-limbs mode.

This full battery was used for habituation purposes prior to the experiment. As administering the full battery (lasting approximately 45 min) was too long for monitoring an immediate effect, we chose to use the 3-Limb combination. This specific combination involves motor inhibition (coupling/decoupling) in the simple condition (recruitment complexity), but the choice condition adds to the processing complexity the selection component which best represents motor inhibition (Boisgontier et al., 2014).

2.4.2. Multilimb simple and choice tasks: the 3-limb combination

Once all four limbs were placed on the tablets, three square plates (representing 3 limbs) on the computer screen turned blue (Fig. 1). In response to this stimulus (which occurred in 2–4 s intervals), participants were asked to release contact with the corresponding tablets as quickly as possible, by lifting the corresponding limbs simultaneously. The reaction time of this task included error corrections. This combination includes the following tasks: a. 2 hands and right foot, b. 2 hands and left foot, c. 2 feet and right hand, and d. 2 feet and left hand. The simple condition was composed of 10 trials for each combination (40 trials altogether). The choice condition was composed of random display of the 4 combinations, each combination was displayed 10 times (40 trials altogether). The duration of the measurement was 12–17 min.

2.4.3. Outcome variables

Motor Reaction Time – The mean elapsed time (in milliseconds) between the onset of the visual stimulus and the release of all correct limbs in each trial. The score is the average of the 40 trials in each of the conditions – simple and choice.

Motor Processing Time – The mean RT in the choice condition minus the mean RT in the simple condition.

2.5. Cognitive performance assessment

2.5.1. Cognitive automatic (Congruent) and complex (incongruent) tasks: vocal stroop test

In order to exclude motor performance from the cognitive assessment, we used the original paper-based version (Stroop, 1935), in which participants are required to read three different subtasks, each including 100 words, as quickly as possible. In the first two subtasks [that represent automatic condition], participants are required to identify names of color patches (Color) and read names of colors printed in black ink (Word). In the third subtask condition (that requires inhibition), color-words are printed in an inconsistent color (for instance the word “red” is printed in blue ink) and participants are required to name the color of the ink rather than reading the written word (Color/Word). The task completion time included error corrections and lasted 2–4.5 min.

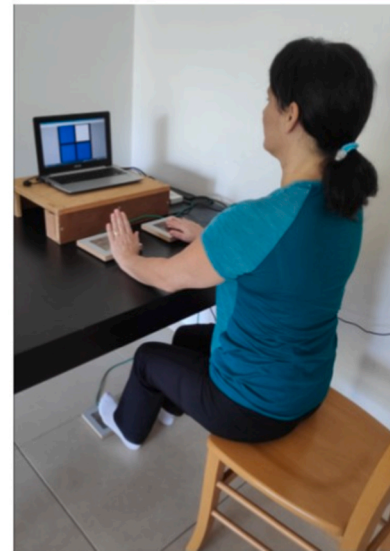
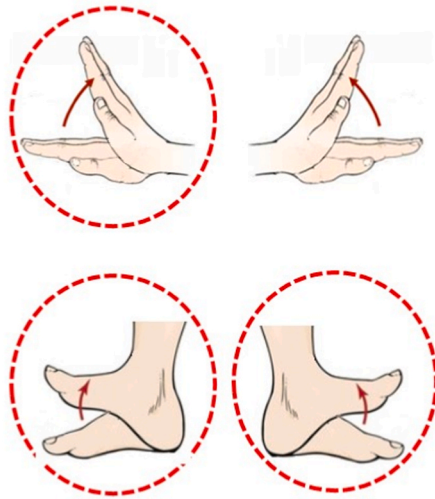


Fig. 1. Demonstration of a 3-Limb task – both feet and left hand. Four squares representing the four limbs are displayed on the computer screen. Once three squares turn blue, the participant is supposed to release contact only with the corresponding limbs of the blue squares.

2.5.2. Outcome variables

Color - the mean *Time* (in seconds) for naming the color of the ink.

Word - the mean *Time* (in seconds) for reading the color.

Color/Word (Incongruent condition) - the mean *Time* (in seconds) for naming the right color (and not reading the color word).

Cognition Processing Time = $\text{Color/Word Time} - [(\text{Word Time} + \text{Color Time}) / 2]$

2.6. Procedure

Participants were asked to meet the experimenter three times on an individual basis:

Day 1: Exercise test and background information. Participants visited one of two exercise laboratories (each located in a different area), signed a written informed consent, provided demographic data, completed the MMSE (Folstein et al., 1975), and were interviewed by a physician about their health as a part of the exercise test procedure. They were then asked to perform the exercise test based on the individual moderate intensity level determined. After this initial meeting, each participant was randomly assigned to either the experimental group or the control group.

Day 2: Establishing a baseline for outcome measures. One to two weeks before the experiment we administered the motor test (full battery) and the cognitive test in a counterbalanced order (participants were randomly allocated to those who started with the motor test and those who started with the cognitive test).

Day 3: Intervention. After 1–2 weeks from the baseline assessment, participants performed the motor (3-Limb) and cognitive inhibition tests (pre-test) in the same order as in baseline. They then completed their assigned treatment (aerobic or control) and performed the same tests again (post-test) within 5 min of completing their treatment. The tests (motor and cognitive) were administered in the same order in all three occasions – baseline, pre, post. Those who were allocated to start with the motor test, started with the motor in all three occasions, etc. The duration of both tests was approximately 15–20 min. Participants in the experimental group began with a slow, 3-min walking warm-up on a motorized treadmill (Vo2 GO 47, SPORTEK, Taiwan). The treadmill speed was then increased (within 2–4 min) until participants reached the target HR (60% of HRR; $\text{HRR} = (\text{HR}_{\text{max}} - \text{HR}_{\text{rest}}) + \text{HR}_{\text{rest}}$). Twenty

minutes of walking at the predetermined target HR was then maintained by adjusting the speed of the treadmill belt, followed by a 3-min cool-down. HR was recorded every 5 min throughout the session by a Polar HR monitor (Polar Electro OY M51). The subjective rating of perceived exertion was measured at 5-min intervals (Borg et al., 1986). Participants in the control group were seated in a quiet room for 26 min and were offered magazines to read.

2.7. Data analysis

In the motor performance we excluded RTs >5,000 msec as they are unreasonable (Boisgontier et al., 2014) and are probably the outcome of a system malfunction. In addition, scores exceeding three standard deviations (SD) from the means were excluded. Altogether, out of 8,640 trials (240 X 36 participants), 341 (3.95%) were excluded – 150 from the experimental group and 191 from the control group.

2.7.1. Baseline analyses

- Independent t-tests and chi-squared tests were used for continuous and categorical variables, respectively, to examine group differences regarding demographic and clinical variables.
- Independent t-tests were performed for assessing whether the order of motor and cognitive tests was a moderator of the differences between groups.
- Two-way ANOVAs with repeated measures (baseline/pre-X experimental/control) were performed on the outcome measures for assessing learning effect (the 3-Limb scores at baseline were generated from the full battery).

2.7.2. 3-Limb task and stroop task, pre to post analyses

- Two-way ANOVAs [Group X Test(pre/post)] with repeated measures on the Test factor were performed on all outcome measures.
- t-tests between groups were administered to the percent of improvement on all outcome measures.
- Pearson correlations were conducted for assessing correlations between improvements in the motor and cognitive variables.

Alpha was set at 0.05, effect sizes are reported as partial eta squared (η^2_p) in the ANOVAs, and as Cohen's d values in the t-tests. All analyses were conducted in JASP (JASP Team, 2021).

3. Results

There were no differences between groups on the demographic, physical and clinical characteristics (see Table 1 above) or between test order (cognitive first vs. motor first) (all p values > .05).

3.1. 3-Limb task and stroop task, pre to post analyses as compared to baseline to pre

Table 2 presents the scores for the 3-Limb tasks (motor performance) and Table 3 – the scores for the Stroop (cognitive performance).

The same trend was revealed for both motor and cognitive tasks. The two-way ANOVAs indicated Test effects (both groups improved), from baseline to the pre-test and from pre to post tests. Although the trends of improvement from pre to post test, in both tasks, was in favor of the experimental group, no interaction was indicated. The statistical data for the two-way ANOVAs for both tasks are described in the S1 Appendix.

In order to inquire more the trend of larger improvement for the experimental group as compared to the control group after the acute exercise, we calculated the percentage of improvement for both groups from baseline to pre-test, and from pre to post exercise. Table 4 presents the percent of improvement from baseline to pretest (representing learning effect), and from pretest to posttest (before and after the acute exercise), for the two groups, as well as statistics, for both motor and cognitive inhibition variables. While the t -tests did not produce significant differences between groups, the effect size (Cohen's d) of the improvement of the exercise group as compared to the control group, was larger and more meaningful for all variables (representing both cognitive and motor inhibition) from pretests to posttests compared to improvements from baseline to pretest (i.e., a learning effect occurred in both groups).

3.2. Exploratory analysis

The trend in favor of the experimental group over the control group on both, cognitive and motor inhibition variables, and the literature discussing the moderating effect of fitness on the influence of acute exercise on cognitive functioning [e.g., (Cui et al., 2020)] led us to explore whether fitness is a moderator of the effect of the exercise on motor and/or cognitive inhibition. For that aim, the scores of the participants in the experimental and control groups were designated as high-fit or low-fit (above and below the 50th percentile, respectively, based on the age and gender fitness tables of the American College of Sports Medicine – ACSM, (see ACSM 2014; Table 4.9, p. 89 for men and p. 92 for women) (American College of Sports Medicine (ACSM), 2014). A three-way ANOVA [Group (exercise/control)] X Fitness (high-fit/low-fit)] X Test(pre/post)] with repeated measures on choice reaction time (motor performance) revealed a Test effect, $F(1,32) = 29.92$, $p < .001$, $\eta^2p = .48$. There was no Group effect, $F(1,32) = 3.22$, $p = .08$, $\eta^2p = .09$, no Fitness effect, $F(1,32) = 0.72$, $p = .40$, $\eta^2p = .02$, no Group $\times$ Fitness interaction, $F(1,32) = 0.12$, $p = .73$, $\eta^2p = .00$, no Group $\times$ Test interaction, $F(1,32) = 0.47$, $p = .50$, $\eta^2p = .01$, and no Fitness $\times$ Test interaction, $F(1,32) = 0.01$, $p = .91$, $\eta^2p = .00$. The three-way interaction was also not significant, $F(1,32) = 3.25$, $p = .08$, $\eta^2p = .09$.

Table 2
Motor performance mean scores (SDs).

Variable		Experimental (n = 20)			Control (n = 16)		
		^a Baseline	Pre	Post	^a Baseline	Pre	Post
Motor Reaction Time (msec)	Simple	631.85 (89.91 $\pm$)	568.92 (82.58 $\pm$)	525.36 (60.16 $\pm$)	640.39 (137.10 $\pm$)	596.11 ($\pm$ 141.16)	570.44 (120.57 $\pm$)
	Choice	1312.29 (222.03 $\pm$)	1178.77 (170.06 $\pm$)	1056.42 (170.57 $\pm$)	1438.30 ($\pm$ 235.01)	1291.81 ($\pm$ 228.12)	1211.04 (213.07 $\pm$)
^b Motor Processing Time (msec)		680.45 (201.77 $\pm$)	609.85 (181.77 $\pm$)	531.06 ($\pm$ 164.35)	797.91 (250.82 $\pm$)	695.70 ($\pm$ 199.93)	640.60 ($\pm$ 159.69)

^a The 3L scores are generated from the full battery.

^b Choice – simple.

However, because of the moderate to large effect size we decided to examine it further and conducted two-way ANOVAs for choice RT for each fitness group separately.

For the low-fit group there were no significant findings (except for the expected pre-post testing effect, $F(1,11) = 9.73$, $p = .01$, $\eta^2p = .47$). However, the two-way ANOVA for the high-fit group revealed a Test effect, $F(1,21) = 23.09$, $p < .001$, $\eta^2p = .52$, and an interaction between Group and Test, $F(1,21) = 4.60$, $p = .04$, $\eta^2p = .18$, with participants in the experimental group improving more than the control group. RT for the high-fit participants in the experimental ($N = 14$) and control groups ($N = 9$) in both simple and choice conditions are presented in Fig. 2. As opposed to motor inhibition, fitness was not a moderator of the effect of acute exercise on cognitive inhibition.

3.3. Correlational analysis

The correlations between the improvement (from pre- to post-test) in Reaction Time in motor performance and the Time in the cognitive performance is presented in Fig. 3. As expected, no correlation was observed on the simple conditions. However, significant correlation was indicated between the inhibition conditions - Choice in the motor inhibition, and Color/Word (incongruent) in the cognitive inhibition with 29% shared variability. Significant correlation was also revealed between the Choice in the motor inhibition, and Color/Word (incongruent) in the Motor Processing Time and the Cognition Processing Time with a shared variability of 15%.

4. Discussion

The aim of the present study was to assess, for the first time, the effect of acute exercise on multi-limb motor inhibition, as compared to cognitive inhibition, in middle-aged adults. We hypothesized that (1) the inhibited response, in both the cognitive and the motor tasks will be altered by exercise, but not the simple response; and (2) motor and cognitive inhibitions are related. Our findings partly support our first hypothesis. Although a trend was revealed in favor of the experimental group in the pretest to posttest improvement on both cognitive and motor tasks, this trend did not reach significance. On the other hand, the effect size (0.32 – 0.58; Table 4) of the group differences in the improvement from pretest to posttest, as opposed to that effect from baseline to pre-test (–0.11 – 0.08), is meaningful, supporting the effect of acute exercise on cognitive and motor inhibition. This trend partly supports previous studies on the effect of acute exercise on cognitive inhibition in middle-aged adults [e.g., (Chang et al., 2019; Netz et al., 2016; Nouchi et al., 2020)]. It may be carefully speculated that had we enrolled more participants to this study, the group differences may have reached significance.

Nevertheless, an exploratory investigation revealed significant differences in favor of the experimental group among the high-fit participants – in movement time during the choice condition of the 3-Limb combination.

This finding reinforces the role of fitness in motor inhibition (Levin et al., 2014; McGregor et al., 2018). It is also in line with previous studies that examined the role of fitness in the effect of acute exercise on executive task-related brain activation, as seen with fMRI (Cui et al.,

Table 3
Cognitive performance (Stroop) mean scores (SDs).

Variable	Experimental			Control		
	Baseline	Pre	Post	Baseline	Pre	Post
Color Time (sec.)	69.80 (15.09)±	60.67 (±13.64)	56.06 (±8.74)	66.17 (13.64±)	60.9 (±12.51)	57.8 (±11.49)
Word Time (sec.)	46.95 (±8.74)	44.54 (±6.78)	43.18 (±5.89)	46.51 (±7.58)	46.09 (±7.39)	45.49 (±8.34)
Color/Word (incongruent) Time (sec.)	107.44 (±24.42)	95.22 (±18.51)	84.05 (±15.76)	102.88 (±20.78)	92.57 (±17.45)	84.74 (±14.35)
^a Cognition Processing time (sec.)	49.07 (22.45±)	42.62 (13.92±)	34.43 (±10.86)	46.55 (16.4±)	39.02 (12.81±)	33.1 (±8.84)

^a Color/Word Time - [(Word Time + Color Time)/2].

Table 4
Percent of improvement for motor and cognitive variables from pre to post-test as compared to improvement from baseline to pre-test.

Variable	Group	Base to Pre				Pre to Post			
		Improvement (%)	Statistic	p	Cohen's d	Improvement (%)	Statistic	p	Cohen's d
^a Motor Reaction Time	Experimental	8.82 (±12.84)	t(34) = -.16	.88	-.05	10.35 (±6.09)	t(34) = 1.74	.09	.58
	control	9.50 (±12.72)				5.81 (±9.48)			
^b Motor Processing Time	Experimental	5.69 (±26.64)	t(34) = -.24	.81	-.08	11.39 (±17.42)	t(34) = .96	.34	.32
	control	7.83 (±26.35)				5.71 (±17.84)			
^c Cognition (incongruent Time)	Experimental	10.36 (±9.10)	t(34) = .24	.82	.08	11.42 (±5.20)	t(34) = 1.65	.11	.55
	control	9.69 (±7.67)				7.73 (±8.10)			
^d Cognition Processing Time	Experimental	13.11 (±18.84)	t(32) ^e = -.32	.75	-.11	18.18 (±10.44)	t(34) = 1.31	.20	.43
	control	15.02 (±15.41)				10.80 (±22.51)			

^a Choice condition.

^b Choice - simple.

^c Stroop color/word.

^d Color/Word Time - [(Word Time + Color Time)/2].

^e Two abnormal.

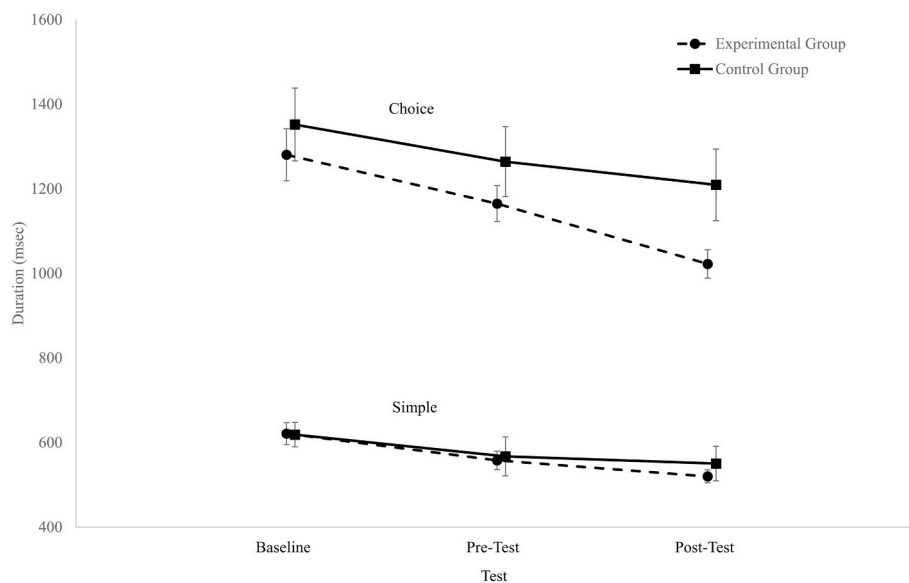


Fig. 2. Reaction Time (RT) in the 3-Limb task for both simple and choice conditions among the high-fit participants. Error bars represent standard errors.

2020; Li et al., 2019; Mehren et al., 2019). For example, Li et al. (Li et al., 2019) suggested that complex cognitive tasks recruit more cognitive resources, which allows individuals with high cardiovascular fitness to perform better in behavioral tasks. As aging is associated with a shift toward accuracy in the speed/accuracy tradeoff (Boisgontier et al., 2016), it is possible that the low-fit individuals in this study, were pre-occupied with the accuracy prioritizing it over speed.

As opposed to the above-mentioned findings of other studies, we did not find effects in the experimental high-fit group in the Stroop test. This finding may be attributed to a combination of moderators: 1. In order to

make a clear distinction between motor and cognitive inhibition, and unlike most studies using Stroop, we applied the vocal version of the Stroop. It is possible that the overall complexity of this task was not sufficiently challenging to mobilize the cognitive resources of high-fit individuals as compared to the motor inhibition task involving 3-Limb coordination tasks. That is, the movement manifestation of the task makes it more complex than the vocal version of the incongruent condition of the Stroop (representing cognitive inhibition). It should be noted that the choice condition of the 3-Limb combination representing motor inhibition in this study, involves in addition to the recruitment

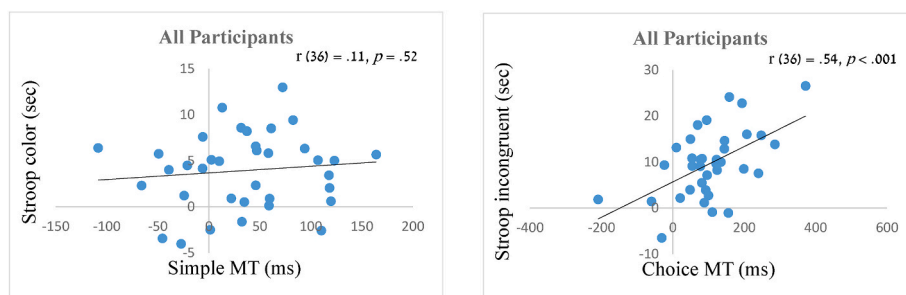


Fig. 3. Scatter plots illustrating correlations between motor and cognitive tasks in pretest-to-posttest improvements. Left: tasks involving low inhibition requirements (simple MT and Stroop color); Right: tasks involving high inhibition requirements (choice MT and Stroop incongruent) RT – Reaction Time; sec – seconds; ms – milliseconds.

complexity (effector-specific networks recruited) also the selection complexity – the coupling/decoupling interactions (Boisgontier et al., 2014). Thus, in support of Li et al. (Li et al., 2019), the high level of complexity of the motor inhibition task, requiring more cognitive-motor resources allowed the participants with high cardiovascular fitness to perform better in the motor inhibition tasks than in the cognitive inhibition task. 2. We used a crude measurement of time in the Stroop – a simple stopper assessing time in seconds, as opposed to the motor test and computerized Stroop applying highly sensitive assessment (milliseconds), possibly enabling the detection of even subtle differences.

As expected, improvements in reaction time were observed in the choice condition (i.e., inhibited condition) but not in the simple condition (Fig. 2). As reported in the literature (Aron, 2007; Diamond, 2013), inhibition tasks involve the suppression of irrelevant responses or stimuli. As such, they require greater effort in terms of information processing – as opposed to simple tasks that require simpler mental responses. Moreover, these inhibition tasks are affected by acute exercise to a greater degree than simple tasks (Levin et al., 2021; Oberste et al., 2019).

Our hypothesis regarding the correlation between motor and cognitive inhibition has been fully supported. While performance in the simple conditions of cognitive and motor functioning did not correlate, the variables representing inhibition were significantly correlated, thereby supporting the argument that cognitive and motor inhibitory functions are mediated by overlapping brain networks (Aron, 2007; Depue et al., 2016; Mirabella, 2014). This pattern was manifested at all stages of the study, regardless of group (experimental or control) or level of fitness. This apparent increase in coupling between cognitive and motor inhibition could be explained by exercise-induced increase in cerebral blood flow to prefrontal cortical regions and motor related areas during and/or after exercise (Giles et al., 2014, 2017; Kojima et al., 2020; Sato et al., 2021). For example, by using near infrared spectroscopy (NIRS), Sato et al. (Sato et al., 2021) reported that a 30-min cycling at 50% of VO_2 max has been found to sustainably increase the levels of oxygenated hemoglobin (O_2Hb) in the supplementary motor area (SMA) and bilateral premotor cortices of healthy adult males. Prefrontal cortical regions including the inferior prefrontal gyrus, dorsolateral prefrontal cortex, SMA and the pre-motor cortices are principal elements of a large-scale network that mediates inhibition [e.g., Aron, 2007; Tan et al., 2019]. As such, it is reasonable to speculate that the simultaneous enhancement in both motor and cognitive inhibition can be attributed, in part, to exercise-induced increase of oxygenation in these regions. The observation that fitness levels appear to contribute to a higher coupling between cognitive and motor inhibition could hint that high-fit group may have better circulation than low-fit and that that oxygenation may be more beneficial for executive-control-related inhibition in the high-fit group. An alternative or a complementary explanation may be that chronic exercise and/or training would result in a higher BDNF synthesis following an acute exercise bout (i.e. compared with untrained participants). Subsequently, more BDNF could be released into the blood

circulation which may, in turn, be absorbed more efficiently by central and/or peripheral tissues where it could induce a cascade of neurotrophic and neuroprotective effects (Knaepen et al., 2010). Yet, another explanation may be that a high level of physical fitness is associated with more efficient response preparation (Renaud et al., 2010) and information processing [e.g., (Williams et al., 2020)], improving both response time and accuracy.

4.1. Limitations

The current study has some important limitations, primarily the low number of participants for our exploratory analyses. Our sample of 36 participants was calculated to provide 80% power to detect a moderate effect in the interaction of a two-way ANOVA with repeated measures on one factor. However, several of our exploratory analyses were conducted with lower sample sizes for which power analyses were not conducted. Some of those included a smaller number of participants and were likely underpowered. This may have led to either type I or type II errors. Specifically, our statistically significant exploratory findings can be spurious [i.e., (Button et al., 2013)]. It should be noted however, that although this can be a spurious finding resulting from a type I error, we were able to detect a significant effect of acute exercise on motor inhibition in high-fit participants. Such findings in an exploratory analysis are meant to generate hypotheses rather than confirm them. As such, we suggest that these exploratory findings can inform researchers for future studies. This limitation is magnified in middle-aged adults as a higher sensitivity of executive function to acute aerobic exercise was found in individuals undergoing developmental changes - preadolescent children and older adults and lower sensitivity in middle-age adults (Ludyga et al., 2016).

Second, as the study do not include any neurophysiological measurements, it was not possible to assess the contribution of GABAergic and glutamatergic signaling pathways, which are known to be affected by acute bout of exercise [e.g., (Singh et al., 2014; Stavrinou & Coxon, 2017); for review see Levin et al. (2021)]. In addition, since the study did not include any measurements of blood flow during and/or after exercise, or BDNF concentration, it was not possible to confirm the direct effect of fitness level on differences in oxygenation or on BDNF level between fit and non-fit individuals. In addition, we were not able to confirm the hypothesis that better circulation or BDNF concentration was a contributing factor to stronger coupling between the two forms of inhibition as proposed above. We propose that future studies would benefit from inclusion of neuroimaging and brain stimulation methods to allow direct associations between beneficial effects of exercise on motor and cognitive inhibition and exercise-induced changes in brain function (e.g., upregulation of glutamatergic activation).

4.2. Conclusions

In conclusion, the main contribution of this study is the examination

of the effect of moderate-intensity acute aerobic exercise on multi-limb motor inhibition in middle-aged adults. Based on the results of this study, a meaningful albeit non-significant trend was seen in favor of the experiment group in their pretest to posttest improvement on both the cognitive and motor tasks. Our result also show that fitness is a moderator of this effect with high-fit participants profit more than low-fit participants. Additionally, based on the rational that cognitive and motor inhibitory functions are mediated by overlapping brain networks we were able to show a relationship between behavioral (performance) tasks representing motor and cognitive inhibition. Furthermore, supporting the potential of acute exercise to improve both cognitive and motor inhibition, this relationship was underscored in the improvement from pre to post-test specifically among the high-fit group. In addition to extending this study to more age groups and to further assess the moderating effect of fitness on the effect of acute exercise on motor and cognitive inhibition, future studies should assess the mediating mechanisms between the neurobiological and behavioral alterations in motor inhibition occurring as a result of acute exercise.

Author contributions

YN and OL conceptualized the study. GZ and SFH analyzed the data. All authors discussed the results. YN wrote the first draft. All authors contributed to improving the draft.

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Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.psychsport.2022.102344>.

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